

**Pokhara University**  
**Faculty of Science and Technology**

Course Code.: CMP 551 (3 Credits)  
Course Title: Artificial Intelligence (3-0-0)  
Nature of the course: Theory  
Level: Master

Full marks: 100  
Pass marks: 60  
Total Lectures: 45 hrs  
Program: MCE/MSc CS

### 1. Course Description

This course is designed to provide a comprehensive introduction to Artificial Intelligence (AI), covering fundamental concepts, algorithms, and applications. Students will explore problem-solving techniques, knowledge representation, deep learning and natural language processing (NLP). The course emphasizes both theoretical foundations and hands-on implementation using modern AI tools and frameworks (Python, TensorFlow, PyTorch, Scikit-Learn).

### 2. General Objectives

The course is designed with the following general objectives:

- To provide the students with the in-depth knowledge of intelligent agents, search algorithms, and knowledge representation techniques.
- To acquaint the students with the skill to design and implement machine learning and deep learning models.
- To equip the students with knowledge to explore real-world applications of AI in areas like reasoning and natural language processing.

### 3. Methods of Instruction

Lecture, Discussion, Readings, Case Studies, Project and Research Works.

### 4. Contents in Detail

| Specific Objectives                                                                         | Contents                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| • Explain the foundation of advanced AI and apply them to build real world AI applications. | <b>1. Foundation of Advanced Artificial Intelligence (7 hrs)</b><br>1.1. Intelligent Agents- Structure, Types and Implementation<br>1.2. Advanced Search Techniques- Bidirectional A* Search, Iterative Deepening A*, Online Search<br>1.3. Local search- Simulated Annealing, Local beam search, Searching in partially observable environments<br>1.4. Games- Adversarial search, Monte Carlo Tree Search, Stochastic Games |
| • Use the knowledge representation techniques and design the reasoning AI systems           | <b>2. Knowledge Representation and Reasoning (7 hrs)</b><br>2.1. First-Order Logic (FOL) and reasoning<br>2.2. Probabilistic reasoning<br>2.2.1. Hidden Markov Models<br>2.2.2. Dynamic Bayesian Networks<br>2.3. Ontological Engineering<br>2.4. Semantic Networks<br>2.5. Description Logic<br>2.6. Fuzzy Logic- Fuzzy Inference Systems (Mamdani, Sugeno and Tsukamoto), Defuzzification techniques                        |



|                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Use the different planning strategies to develop the intelligent agents</li> </ul>       | <b>3. Planning and Decision Making (6 hrs)</b><br>3.1. STRIPS and PDDL<br>3.2. Hierarchical Planning<br>3.3. Probabilistic Planning<br>3.4. Multi-agent Planning<br>3.5. Utility Theory<br>3.5.1. Utility Function<br>3.5.2. Multi-attribute utility Function<br>3.6. Decision Networks<br>3.6.1. Sequential Decision Problems<br>3.6.2. Algorithms for MDPs<br>3.6.3. Partial Observable MDP |
| <ul style="list-style-type: none"> <li>• Solve the problems that can be formulated as constraint satisfaction problems</li> </ul> | <b>4. Constraint Satisfaction Problems (5 hrs)</b><br>4.1. Defining CSPs<br>4.2. Constraint Propagation<br>4.3. Inference in CSPs<br>4.4. Backtracking Search for CSPs                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"> <li>• Apply the CNN and RNN for various AI applications</li> </ul>                             | <b>5. Neural Network and Deep Learning (10 hrs)</b><br>5.1. Perceptron, Multi-Layer Perceptron (MLP) and Backpropagation<br>5.2. CNN (Convolutional Neural Networks)- Zero-shot, Few-shot learning, Graph CNN<br>5.3. RNN (Recurrent Neural Networks)<br>5.4. Attention Mechanisms<br>5.5. Transformers<br>5.6. Graph Attention Networks<br>5.7. Transfer Learning                            |
| <ul style="list-style-type: none"> <li>• Design NLP systems</li> </ul>                                                            | <b>6. Natural Language Processing (10 hrs)</b><br>6.1. Language Models<br>6.2. Part-of-speech (POS) Tagging<br>6.3. Grammar and Parsing<br>6.4. Complications of Real Natural Language<br>6.5. Word Embeddings<br>6.6. RNN for NLP<br>6.7. LSTMs for NLP<br>6.8. Sequence-to-Sequence Models<br>6.9. The Transformer Architecture- BERT, GPT, Attention                                       |

## 5. Evaluation System and Students' Responsibilities

### Evaluation System

The internal evaluation of a student may consist of assignments, attendance, internal assessment, project and research works etc. The internal evaluation scheme for this course is as follows (but not limited to):

| Internal Evaluation              | Weight | Marks | External Evaluation      | Marks |
|----------------------------------|--------|-------|--------------------------|-------|
| Attendance & Class Participation | 10%    | 60    | Semester-End examination | 40    |

|                                  |     |           |                       |           |
|----------------------------------|-----|-----------|-----------------------|-----------|
| Assignments                      | 20% |           |                       |           |
| Internal Assessment              | 40% |           |                       |           |
| Research/Project/Case Study Work | 30% |           |                       |           |
| <b>Total Internal</b>            |     | <b>60</b> | <b>Total External</b> | <b>40</b> |
| Full Marks: 60 + 40 = 100        |     |           |                       |           |

### Students' Responsibilities

Each student must secure at least 60% marks separately in internal evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

### 6. Reference Books

1. Russell, S. J., & Norvig, P. (2016). *Artificial intelligence: a modern approach*. Pearson.
2. Goodfellow, A. I. (2016). *Deep learning-ian goodfellow, yoshua bengio, aaron courville-google books*.
3. Bishop, C. M. (2006). *Pattern recognition and machine learning* (Vol. 4, No. 4, p. 738). New York: Springer.
4. Aurélien, G. (2017). Hands-on machine learning with scikit-learn & tensor flow. *Geron Aurelien*, 134, 145-150.
5. Jurafsky, D., & Martin, J. H. *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*.
6. Sutton, R. S., & Barto, A. G. (1998). *Reinforcement learning: An introduction* (Vol. 1, No. 1, pp. 9-11). Cambridge: MIT press.
7. Koller, D., & Friedman, N. (2009). *Probabilistic graphical models: principles and techniques*. MIT press.